Mr. Davies Homework task 3

Due date

Sun 16 May 4:00pm

1. Write $3^4 \times 3^{10}$ in simplest index form.

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2. Write $5^{10} \div 5^5$ in simplest index form.

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3. Write $\left(3^7\right)^5$ in simplest index form.

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4. Simplify the expression $2t + 4 \times 8t$.

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5. Simplify the expression $\frac{16x}{3x + 5x}$.

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6. Simplify the expression $\frac{20s}{4s} \times 6t$.

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7. Expand the expression $5(s + 7)$.

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8. Expand the expression $(y + 8) \times 7$.

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9. Expand the expression $-(y+2)$.

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10. Expand the expression $-8(-2t-3)$.

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11. Expand the expression $p(q+4)$.

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12. Expand the expression $r(r+7)$.

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13. Expand the expression $2f(f-7g-6h)$.

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14. Expand and simplify the expression $3(6x-5y)-2(9y+4x)$.

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15. Expand and simplify the expression $x(x+4)+8(x+9)$.

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16. What is the highest common factor of 2 and $6x$?

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17. What is the highest common factor of $35x$ and 14?

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18. Factorise $3w + 15$.

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19. Factorise $40c - 35$.

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20. Factorise $-8w^2 + 3w^2y$.

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21. What is the highest common factor of $24m$, $12n$ and $18q$?

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22. Complete the factorisation of $3m - 6n + 12p$:

$$3m - 6n + 12p = 3\left(\boxed{} - \boxed{} + \boxed{}\right)$$